Krishna's Technical Clarifications — Architecture Team Assessment

LocalAI TV · v1.3 Review · For Cross-Review by AI Eng Team + Krishna

To: AI Engineering Team (Sheikh Sameer, Gnana Rajnan), Nagarjuna Reddy (CTO) **and** Krishna (DevOps) — for review & comment **From:** LocalAI TV Architecture Team **Subject:** Krishna's 9 technical clarifications on v1.3 — reproduced in full, with a point-by-point assessment (plain-language) **Date:** 16 May 2026 **Status:** 🟡 Assessment only — **no action taken, v1.3 unchanged, nothing implemented.** Circulated so the AI team and Krishna can comment; founder then decides.

> **How to read this.** Part 1 is Krishna's submission reproduced **exactly as received, unedited**. Part 2 is the architecture team's assessment in plain language. Part 3 is the overall verdict. Part 4 cross-checks against the other reviews. Part 5 states the v1.3 impact (none now — deferred items only).

Part 1 — Krishna's Original Submission (verbatim, unedited)

> *Reproduced exactly as supplied in Krishna's PDF "Technical Clarifications Required." No wording changed.*

Technical Clarifications Required — Architecture & Infrastructure Review — Open Points for Sign-off Raised by: Krishan · **To:** Mohan, Arjun · **Date:** 16 May 2026

The following points need clarification before we proceed. Kindly review each and acknowledge with your decision or guidance. These are the main open items from the current architecture document.

1. Direct DB Access vs API for Scheduler *Context:* Admin Dashboard scheduling needs a defined integration path with the AI-side scheduler. *Document says:* API-based integration is the recommendation. *Need confirmation:* Should the Admin Dashboard call the AI-side scheduling APIs directly? Please acknowledge if API-based integration is the agreed approach (vs. direct DB access).

2. Cloudflare R2 vs AWS S3 Confusion *Issue:* Some sections mention Cloudflare R2 uploads, but the overall architecture is finalized on AWS S3 (Mumbai). The document currently mixes both. *Need clarification:* Is R2 only a temporary upload-staging layer, or is it fully removed from the design? A single source of truth is needed.

3. Supabase Free Tier — Production Reality Check *Issue:* Document assumes Supabase free tier initially. At projected scale this must support pg_boss, RLS, analytics, cron, and audit logs. *Need:* Actual production sizing for the free-tier plan — a realistic check on whether the free tier can survive production load, or what paid tier is required.

4. AI Pipeline DB Ownership Boundary *Not defined:* Ownership and process boundaries between teams are unclear. *Need decisions on:* Who owns migrations; schema versioning; rollback strategy; and cross-team release coordination.

5. 3,000-Channel Projection — Real Capacity Planning *Issue:* Document provides cost estimates but no infrastructure capacity model. *Missing:* S3 request-rate calculations; RabbitMQ throughput; Redis memory sizing; GPU worker concurrency math; FFmpeg NVENC parallelization limits; API rate-limit projections. *Need:* Actual infra capacity modeling backing the 3,000-channel projection.

6. YouTube API Quota Problem (Critical) *Issue:* Document itself notes default quota supports only ~6 uploads/day/project, but the target is 600,000 uploads/day. This is currently unsolved operationally. *Need:* Either a multi-project quota orchestration strategy, OR an alternative live architecture. This is a blocking item.

7. No CI/CD Architecture Defined *Missing:* Deployment pipeline and blue/green deployment strategy are not defined. *Need:* A defined CI/CD architecture and release/rollback flow.

8. No Media Transcoding Profiles Defined *Missing:* Bitrate ladder; adaptive streaming strategy; HLS/DASH (MPEG-DASH) packaging; mobile bandwidth optimization. Currently only FFmpeg is mentioned. *Need:* Defined transcoding profiles and an adaptive streaming/packaging strategy.

9. Analytics System Underspecified *Referenced:* Analytics section references drill-down and dashboards. *Missing:* OLAP strategy; aggregation pipelines; retention policy; real-time vs. batch analytics; warehouse design. *Context:* OLAP is designed for high-speed, complex analysis over massive historical data. Where a standard DB handles day-to-day operations, an OLAP system answers deep aggregate questions over months/years. Example cube for video-processing data: Time (days/weeks/months/quarters) x Geography (districts/states/countries) x Content Category (news/birthdays/events/classifieds). *Need:* A defined analytics/OLAP architecture with aggregation, retention, and warehouse design.

Please acknowledge each point with your decision or guidance so we can finalize the architecture document.

Part 2 — Architecture Team Assessment

Every point checked against the actual v1.3 text and the existing decision trail (the Gyan DB response, gap analysis, Production Clarifications, and the AI team's Architecture Review).

Summary table

#	Topic	Genuine?	Already resolved / covered?	Verdict
1	Scheduler: API vs direct DB	Valid question	**Decided** — Option A (API), founder-approved	Already answered
2	R2 vs S3 confusion	Valid	**Decided 3×** — S3 only; in v1.4 cleanup	Already answered
3	Supabase free tier won't survive prod	✅ **Sound**	Not fully costed in v1.3	**ADOPT**
4	DB ownership: migration/rollback process	✅ **Sound**	Ownership defined; *process* not	**ADOPT**
5	3,000-ch capacity model	Principle valid	AI team's review **did the math**	Already covered
6	YouTube quota (critical)	Yes	AI team's review **= hard blocker, Phase-1 negotiation**	Already identified
7	No CI/CD defined	✅ **Sound**	Genuinely missing	**ADOPT**
8	No transcoding/ABR profiles	Yes	AI team's review = open clarification C3	Already identified
9	Analytics OLAP underspecified	Partial	`content_filter_counts` mitigates; OLAP premature	Mostly covered

Detail

1 — Scheduler API vs direct DB · Already answered. The founder approved **Option A (API-based)** in the Gyan DB-spec response. Answer to Krishna: *yes, API-based is the agreed approach; the Admin Dashboard calls the AI-side scheduling API — not direct DB access.* No open item.

2 — R2 vs S3 · Already answered. Correct observation, but identified and resolved three times already (Abishek #3, gap analysis, Production Clarifications). Decision: **R2 is fully removed; AWS S3 (Mumbai, `ap-south-1`) is the single source of truth; Cloudflare = CDN/security only.** Already queued in the v1.4 cleanup (Addendum A). No new action.

3 — Supabase tier · ADOPT. Technically sound. v1.3's cost view optimistically treats Supabase/queue as ₹0. At production scale (pg_boss + RLS + analytics + cron + audit logs) the free tier is not viable — this aligns with the AI team's pg_boss-at-scale concern. **Adopt:** Supabase must be a sized **paid tier**; fold into the cost re-baseline. (Already reflected in the Founder-Requirements doc as Supabase Pro — Krishna's point reinforces it.)

4 — DB ownership/process boundary · ADOPT. The two-database **federation model already defines *who owns which database*** (Admin team ↔ AI team, linked by `content_id`). What is **not** defined is the **operational process**: migration ownership, schema versioning, rollback, cross-team release coordination. That is a real, correctly DevOps-shaped gap. **Adopt** as a Phase 1 process-definition item.

5 — Capacity model · Already covered. Valid in principle, but the AI team's Architecture Review already performed the capacity math (GPU 70–100 instances at peak; RabbitMQ ≈7 msg/s — "never the bottleneck"; CDN 25× under). Krishna lists categories to model; the AI team already modelled them. Agreed plan stands: **re-baseline Phase 3 capacity after Phase 1 with real data.** Nothing new.

6 — YouTube quota · Already identified. Correctly flagged as critical/blocking — and it already is, in the Architecture Review (hard blocker; begin quota negotiation during Phase 1; Phase 3 viability depends on it). Independent corroboration, not a new item.

7 — CI/CD · ADOPT. Genuinely absent from v1.3. A defined deployment pipeline + release/rollback flow is a legitimate gap and correctly DevOps-shaped. **Adopt** as a Phase 1 item (CI/CD + rollback strategy).

8 — Transcoding/ABR · Already identified. Same point as the Architecture Review's adaptive-bitrate-ladder recommendation, already an open clarification (C3: does a 3-rung HLS ladder fit the GPU budget). Corroboration, not new.

9 — Analytics/OLAP · Mostly covered. The valid kernel — at scale, analytics must be pre-aggregated, not live-queried — is **already designed** via `content_filter_counts` (precomputed counts, explicitly "no live queries at scale"). A full OLAP/warehouse design is **premature** for the current stage and reads as generic. **Adopt only the kernel:** define analytics aggregation + retention as a **Phase 2/3 design item** (design later, not now).

Part 3 — Overall: Is the Review Technically Sound?

A fair, evidence-based answer to the founder's question.

- **Mixed — but not noise, and not to be dismissed.**
- **3 of 9 are genuinely worth adopting** (#3 Supabase sizing, #4 DB lifecycle process, #7 CI/CD). These fill an **operations/process gap that neither Abishek nor the AI team's pipeline review covered** — which is exactly what a DevOps engineer is supposed to catch. Real value.
- **2 of 9 are already decided** (#1, #2) — Krishna re-raised them because he reviewed v1.3 **without the decision trail** (the response docs that already resolved them).
- **3 of 9 were already identified by the AI team's Architecture Review** (#5, #6, #8) — useful independent corroboration, not new findings.
- **1 of 9 is mostly covered and over-framed** (#9 — the textbook OLAP definition and example cube read as generic AI-explainer content; the real need is already mitigated by precomputed counts).

Signs of shallow / AI-assisted review are present (a pasted textbook OLAP definition; a capacity "checklist" with no math; re-raising settled items). **But** he independently surfaced three real, cleanly-scoped operational gaps the deeper engineers did not — so the fair conclusion is: **Krishna has genuine DevOps instincts on the operations/process layer; his reviews need grounding against the existing decision trail before they reach the founder** (which is what this assessment does). Use him for the ops/process layer; filter his points against prior decisions first.

This is not a criticism of effort — #3/#4/#7 are solid, useful catches.

Part 4 — Cross-Check Against the Other Reviews

Krishna point	Overlaps with	Net
#2 (R2/S3), #6 (YouTube), #8 (ABR), #5 (capacity)	Abishek #3 / Architecture Review	Already caught — confirms they're real
#1 (scheduler)	Gyan DB response (Option A decided)	Already answered
#3, #4, #7	**Not covered by anyone else**	**Krishna's unique, genuine contribution**

Three independent reviews now converge on the same core issues — strong confirmation the architecture is sound and the known gaps are correctly identified. Krishna adds the ops/process layer.

Part 5 — Impact on v1.3 (Founder's question: anything to implement?)

Nothing is implemented. v1.3 is unchanged. The gate discipline is intact.

The three **ADOPT** items (#3 Supabase tier, #4 DB lifecycle process, #7 CI/CD) are **recorded for the Phase 1 backlog / the deferred v1.4 Addendum A** — to be applied in one pass **only after the Phase 0 Validation Gate clears**, exactly like every other accepted correction. They do **not** change Phase 0 and do **not** start now.

- Addendum A additions: **#12** Supabase paid-tier sizing · **#13** DB migration/versioning/rollback/release-coordination process · **#14** CI/CD + rollback architecture.
- Everything else: already answered or already on the list.

So — to the founder's question: **no, nothing to implement in v1.3 now.** Three items recorded for later (post-gate), nothing more.

Sign-off (for cross-review)

Party	Asked to	Status
Sameer & Gnana / Nagarjuna	Review Krishna's points + this assessment; agree / add	☐ Pending
Krishna	Review this assessment; respond on #3, #4, #7 (the adopted items)	☐ Pending
Founder — Koneti Mohan Reddy	Note: 3 adopted → Phase 1 backlog; rest answered. No v1.3 change now	☐ Pending

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Assessment only. v1.3 unchanged. Adopted items deferred to the post-gate v1.4 pass. No coding until scope is frozen and the founder lifts the gate.